Application of Multi-scale Path Planning for Quadruped Robots in Special Environments

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***Abstract*—In recent years, autonomous navigation technologies for quadruped robots have achieved significant advancements in structured road and indoor environments. However, in special environments such as underground coal mines and coal preparation plants, challenges persist due to their semi-structured nature. This paper introduces an existing Multi-scale Path Planning (MPP) method and applies it to the navigation tasks of quadruped robots in special environments. The MPP method integrates a global planner, a mid-layer planner, and a local planner, enabling the generation of global paths while effectively addressing obstacles of varying scales. The global planner provides overall navigation guidance based on road connectivity, the mid-layer planner constructs a cost map to perceive large-scale obstacles and generate intermediate paths, and the local planner utilizes the Terrain-considered Timed Elastic Band (TC-TEB) algorithm to optimize local trajectories by incorporating terrain-speed constraints. This ensures vertical stability and path safety. Experimental results demonstrate that the MPP method achieves efficient and stable navigation in both simulated and real-world special environments.**

***Index Terms*—Quadruped Robots, Multi-scale Path Planning, Special Environments, Autonomous Navigation**

I. Introduction

The primary objective of autonomous navigation for quadruped robots is to determine and optimize their paths to reach target points safely and efficiently. In structured indoor and road environments, autonomous navigation technologies have achieved significant progress and widespread application. In recent years, some researchers have advanced techniques for navigating quadruped robots in specialized environments, such as underground coal mines and coal preparation plants. For example, Xie et al. [1] introduced the Circular Accessible Depth (CAD) representation for scene perception in semi-structured environments. Similarly, Jian et al. [2] developed the Plane-Fitting Uneven Terrain Navigation (PUTN) method, which evaluates terrain traversability through plane fitting, demonstrating notable navigation performance in these challenging environments.

Specialized environments, such as underground mines, coal preparation plants, and other semi-structured areas, typically combine structured scenes (e.g., tunnels) with unstructured ones (e.g., regions with obstacles or irregular terrain). Navigating these environments is highly challenging due to the presence of unexpected obstacles of various shapes and sizes. Current methods are primarily designed to handle obstacles of conventional scales but often struggle with the frequent occurrence of large-scale or small-scale unstructured obstacles in complex environments.

To address these challenges, this study introduces and applies a multiscale path planning (MPP) method to navigate quadruped robots in specialized environments. MPP integrates a global planner for directional guidance, a mid-layer planner to avoid large-scale obstacles, and a local planner that incorporates the Terrain-Consistered Timed Elastic Band (TC-TEB) to optimize speed and navigate small-scale obstacles effectively.

This is shown in Fig. 1. The methodology of the Multiscale Path Planning (MPP) framework integrates global, mid-layer, and local planning to address the challenges of navigating quadruped robots in complex environments. [3] The global planner establishes a target point for path planning and performs a graph search over a road topology map to generate a globally optimal path with minimal cost, serving as directional guidance for subsequent planning stages. The mid-layer planner incorporates the Circular Accessible Depth (CAD) representation into its perception module, creating a mid-layer cost map to identify and avoid large-scale obstacles. By combining the global path with the mid-layer cost map, the MPP effectively generates paths that bypass large-scale obstacles. For local planning, terrain traversability plays a critical role in speed control when navigating uneven terrain. While the original Timed Elastic Band (TEB) algorithm [4] considers only obstacle information, it overlooks speed optimization. To address this, the local planner analyzes terrain characteristics—such as roughness, slope, and edges—using elevation

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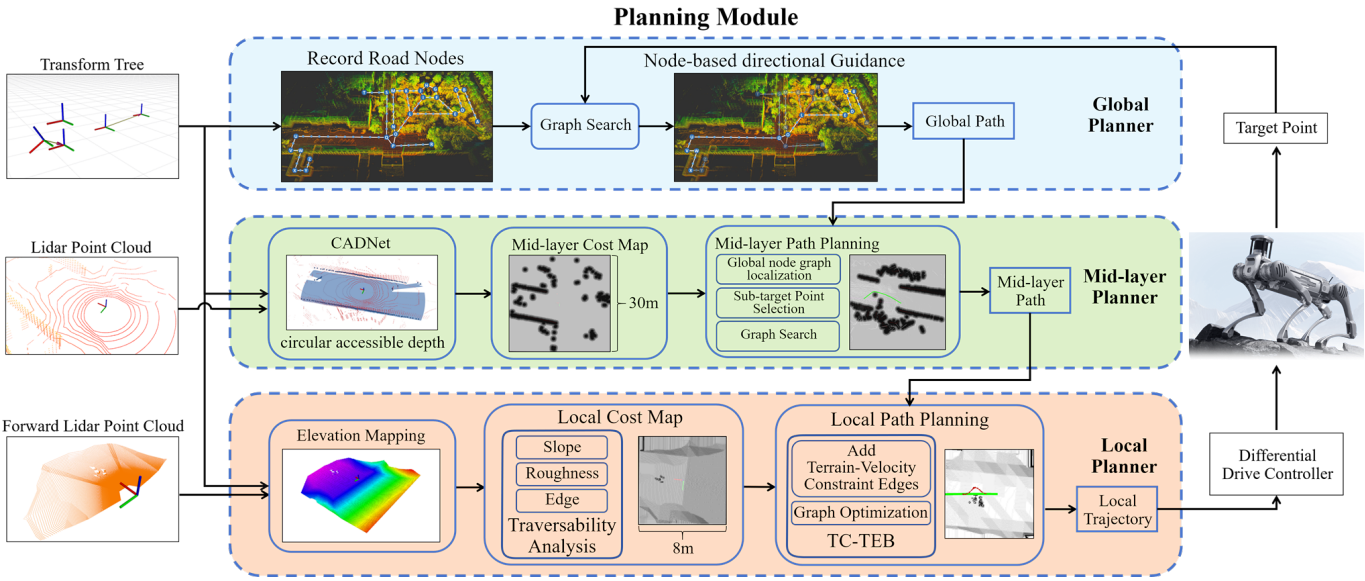


Fig. 1. Overview of the ours method. The method which uses LiDAR point cloud data as input and employs a transformation tree to establish relationships between different coordinate systems. The overall framework is illustrated on the right, from top to bottom: the global planner determines the optimal connections between road nodes, where blue circles represent primary nodes and blue dots indicate secondary nodes. The midlevel planner generates a midlevel path by integrating the global path with a cost map constructed from CADNet predictions, effectively avoiding large-scale obstacles. Finally, the local planner performs a traversability analysis based on elevation features to create a traversability cost map. Using the TC-TEB algorithm, it optimizes trajectory planning and generates motion control inputs for the quadruped robot, enabling precise path tracking and robust navigation.

maps. The developed Terrain-Considered Timed Elastic Band (TC-TEB) algorithm enables the robot to avoid small-scale obstacles while optimizing speed control on traversable but high-cost terrain (e.g., speed bumps or highly uneven surfaces), ensuring smoother and safer navigation.

The contributions of this study are summarized as follows:

- Introduction and application of the MPP method for quadruped robot navigation in specialized environments, integrating global, mid-layer, and local planners. The method's effectiveness is validated in both simulated and real-world environments.
- The development of a novel mid-layer planner that integrates the traversable area generated by CAD with subgoal selection for path planning, effectively addressing large-scale obstacles.
- Introduction of the Terrain-Considered Timed Elastic Band (TC-TEB) algorithm, which incorporates terrain traversability to optimize speed control, ensuring effective navigation on challenging terrain.

II. Related Work and Method

The method employed in this study is a navigation approach based on Multi-Scale Path Planning (MPP), which also falls under the category of LiDAR-based navigation methods. This section provides a concise overview of related works in these two domains.

A. LiDAR-Based Methods for Navigation

LiDAR-based navigation methods can be categorized based on the type of environmental representation used: 2D representation, 2.5D representation, and 3D representation.

1) 2D Representation: Occupancy Grid Maps (OGMs) represent environments as 2D grids, utilizing probabilities to indicate traversability. A method was introduced to construct global and local cost maps by processing laser scans or point clouds. This approach was further refined by developing hierarchical cost maps to better manage different types of obstacles. Advanced neural network techniques, such as Recurrent Neural Networks (RNNs), have been applied to predict dynamic OGMs from 2D laser scans [5]. Additionally, ray projection techniques have been employed to segment non-ground points and train models to forecast future occupancy states. [6]

2) 2.5D Representation: To incorporate vertical information, 2.5D representations extend 2D maps by adding elevation data. A method was developed to create 2.5D navigation maps that capture both horizontal and vertical obstacles, enhancing path feasibility assessments. Another approach combined 2D grid maps with geometric 3D features to build hybrid 2.5D maps suitable for dynamic environments. Elevation mapping techniques address drift errors from the robot's perspective, while methods using GPUs have been introduced to accelerate the process. Sparse LiDAR data has also been used for traversability analysis, and multimodal fusion has been employed to provide configurable 2.5D environmental representations. [7] However, due to their limited perception range, elevation maps are primarily used as local perception modules.

3) 3D Representation: For more detailed and comprehensive mapping, 3D representations use volumetric grids or point clouds. A voxel-based navigation framework has been proposed, which is adaptable to various 3D environments. 3D

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voxel grids have been utilized for unstructured terrains, though challenges remain in large-scale modeling and obstacle representation. To address these issues, a system was developed that avoids full environment modeling but faces difficulties in unstructured scenarios. Additionally, methods using 3D occupancy maps provide detailed evaluations of traversability but incur significant memory and computational costs, which can negatively impact the performance of robots, especially in large outdoor environments. [8]

Each representation type has unique advantages and limitations, making their selection highly dependent on the specific navigation requirements and environmental constraints.

B. Multiscale Path Planning for Navigation

This subsection presents a multi-scale navigation framework designed to process LiDAR point cloud data as input. The framework incorporates three hierarchical planning modules—global, regional, and local—seamlessly integrating them to iteratively enhance the navigation efficiency and robustness of the quadruped robot.

1) Point Cloud Localization and Registration

Point cloud maps provide a unified representation for both indoor and outdoor navigation, capturing precise three-dimensional environmental structures.

The Normal Distribution Transform (NDT) algorithm is employed for point cloud registration, minimizing the matching error function in (1):

$$\text{score} = \sum_i \exp\left(-\frac{1}{2}(x_i' - \mu_i)^T \Sigma_i^{-1}(x_i' - \mu_i)\right) \quad (1)$$

where μ and Σ represent the mean and covariance matrix, respectively, describing the distribution of point cloud data within a grid.

To address the limitations of NDT in sparse feature environments, a coarse-to-fine registration method is proposed: The FAST-LIO algorithm integrates LiDAR and IMU data to provide a more accurate initial pose estimation.

The initial coarse pose T_t^{guess} is computed as (2):

$$T_t^{\text{guess}} = T^{\text{map}\to\text{init}} \cdot T_t^{\text{init}\to\text{body}} \quad (2)$$

2) Global Path Planning

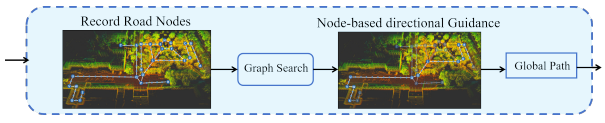


Fig. 2. Overview of Global Planner.

As shown in Fig. 2. A road topology graph is constructed based on the point cloud map, marking both primary and secondary nodes while recording connectivity. Graph search algorithms, such as Dijkstra's algorithm, are used to compute the optimal path between nodes, where the edge weights are defined by the Euclidean distance, like (3):

$$\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (3)$$

Secondary nodes are inserted between primary nodes to enhance mapping accuracy, ensuring that both the target and current positions are accurately represented within the topology graph.

3) Regional Path Planning

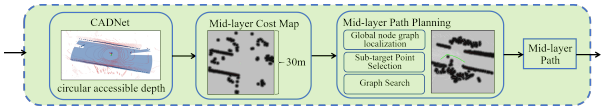


Fig. 3. Overview of Mid-layer Planner.

As shown in Fig. 3. Regional path planning supplements global planning by incorporating obstacle information through cost maps constructed from Circular Accessible Depth (CAD): CAD transforms point cloud data from polar coordinates to Cartesian coordinates as (4):

$$\begin{aligned} x_b &= r\cos\phi \\ y_b &= r\sin\phi \end{aligned} \quad (4)$$

Dynamic planning algorithms, such as D*, are employed to adaptively optimize the path as the environment changes.

4) Local Path Planning

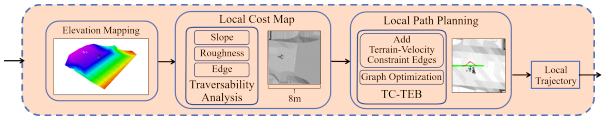


Fig. 4. Overview of Local Planner.

As shown in Fig. 4. Local path planning addresses limitations in regional planning by leveraging elevation maps for terrain traversability analysis, ensuring safe and smooth vehicle trajectories. - The optimization goal for local path planning is formulated as (5):

$$\min \sum \text{Cost}(x, y, z) + \text{TEB constraints} \quad (5)$$

where $\text{Cost}(x, y, z)$ accounts for terrain undulations, obstacles, and other factors, while TEB ensures trajectory smoothness and dynamic feasibility.

The multi-scale planning approach effectively balances computational efficiency and path quality by combining global guidance, regional obstacle avoidance, and precise local trajectory optimization.

III. Experiments

A. Experimental Setup and Environment

The goal of this experiment is to evaluate the proposed *Multi-Scale Path Planning (MPP)* method in specialized environments such as coal mines and coal preparation plants. The experiment consists of both simulation and real-world tests to verify the performance of the proposed method under different conditions.

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TABLE I
QUANTITATIVE RESULTS OF NAVIGATION EXPERIMENTS

Method	Collision Rate (%)	Success Rate (%)	Average Speed (m/s)	Postural Stability (m)
MPP Method	**0**	**98.5**	1.0	**0.03**
CAD Method	3.1	92.3	0.9	0.12
Move_base	5.4	89.7	**1.2**	0.10

1) Environment Description: The experimental scenarios are based on typical environments found in coal mines and coal preparation plants, including the following:

- **Underground Coal Mine Tunnels:** Complex spatial structures with narrow paths and irregular obstacles.
- **Coal Preparation Plant Ramps:** Sloped terrain with stones and uneven surfaces.
- **Mining Transport Roads:** Roads with speed bumps, potholes, rocks, and uneven terrain.

All experimental environments were created in a simulation platform and validated in real-world conditions.

2) Simulation Experiment Setup: In the simulation environment, we used ROS and Gazebo for simulation. Various obstacles, such as small stones, potholes, and large-scale obstacles, were included in the scenarios. Different obstacle densities and placements were configured to evaluate the performance of the MPP method in complex environments.

3) Real-World Experiment Setup: The real-world experiments were conducted in a selected coal mining area using an autonomous navigation platform equipped with the MPP method (e.g., unmanned ground vehicles or quadruped robots). The experimental setup included high-precision LiDAR and IMU sensors for real-time localization and obstacle detection.

The specific experimental environment is shown in Fig. 5.



(a) Simulation Environment (b) Real-world Environment

Fig. 5. The schematic diagram of the virtual and real environments.

B. Experimental Metrics

To comprehensively evaluate the performance of the proposed method, the following four key metrics were used:

- **Collision Rate:** Collisions were categorized into large-scale obstacles, small-scale obstacles, and others. The collision rate is defined as the ratio of the number of collisions to the total number of tests.
- **Success Rate:** Defined as the ratio of the number of successful tests where the vehicle reaches its navigation target without collisions.
- **Average Speed:** Calculated as the mean velocity of the quadruped robot across all successful navigation tests.
- **Postural Stability:** Measured as the standard deviation of the robot's pitch and roll angles, reflecting the robot's ability to maintain balance and posture during navigation, particularly in uneven or challenging terrain.

C. Experimental Results and Analysis

The MPP method, along with two baseline methods, *CAD* and *Move_base*, were compared in both simulation and real-world experiments. Each method was tested in the same experimental environments, and the results are presented in Table I.

From the experimental results, it is evident that the proposed MPP method outperforms all other methods in terms of the lowest collision rate and highest success rate. Additionally, the MPP method exhibits an average speed that meets the required standard, being higher than CAD and slightly lower than Move Base, while also demonstrating better postural stability compared to both Move Base and CAD, indicating its reliable stability in complex and irregular terrains.

IV. DISCUSSION AND CONCLUSION

The experimental results from both simulations and real-world tests validate the effectiveness of the proposed Multiscale Path Planning (MPP) method for quadruped robot navigation in specialized environments, such as coal mines and coal preparation plants. Compared to traditional methods, MPP shows superior performance in navigating narrow spaces, avoiding small-scale obstacles, and traversing uneven terrain, while maintaining stability and speed. The integration of global, mid-layer, and local planners, along with the Terrain-Considered Timed Elastic Band (TC-TEB) algorithm, enhances navigation robustness in complex environments. Future work will focus on optimizing real-time performance and expanding the method's applicability to other challenging environments.

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